Graph and Geometric Deep Learning for Predicting Drug Safety Properties

Mid-Term Progress Report

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Abstract

We study graph neural networks for drug safety prediction. The proposal asked whether GNNs outperform classical fingerprint baselines on a pharmaceutical task. We narrowed the application from drug-target binding to drug toxicity, built a five-level baseline ladder, and ran four GNN architectures in two configurations across six toxicity datasets and twelve Tox21 endpoints. GIN is the strongest GNN, and after a v2 upgrade (richer atom features, higher capacity, class-weighted loss) it matches fingerprints on the small or severely imbalanced classification tasks but trails on the balanced ones and on LD50 regression. Loss-side imbalance fixes inside the GNN training outperform data-side fixes (rejection sampling, SMOTE) on the classical baselines. The final phase hardens the GNN (bond-level edge features, scaffold splits with multiple seeds, focal loss for the one architecture that regressed under aggressive class weights) and takes the hardened GIN drug encoder to drug-target interaction on DAVIS, the regime where the protein partner cannot be Morgan-fingerprinted. A 3D-geometric GNN on LD50 (SchNet, Level 4) is a stretch goal.

1 Introduction

1.1 Research question

Drug development is dominated by attrition. Most candidate molecules fail in clinical trials, and a substantial fraction of those failures are toxicity-driven. Predicting safety from molecular structure alone has direct economic and ethical value, and the same modelling tools transfer to drug-target interaction (DTI), where the original proposal aimed.

The proposal asked: *do graph-based and geometry-aware representations of molecules provide a meaningful advantage over classical descriptor-based baselines on a pharmaceutical prediction task?* The task was originally drug-target binding for a single protein. We refined it to drug toxicity because the Therapeutics Data Commons benchmarks [1] provide mature toxicity datasets with standardised splits and the methodological question transfers cleanly. DTI on DAVIS and KIBA is already loaded in the pipeline and is the planned target of the final phase (§6). This report is a mid-term checkpoint: it documents the exploration, defines the problem precisely, and motivates the final-phase plan.

Going in, we expect graph representations to preserve more relevant structural information than fixed-length fingerprints, particularly around ring systems, hydrogen bonding, and stereochemistry. Classical fingerprint baselines should be hard to beat on small datasets and on highly imbalanced tasks, where flexible models risk overfitting the minority class. Whether a GNN advantage emerges on DTI, where the protein cannot be reduced to a fingerprint, is the open question.

1.2 Baseline hierarchy

Following the structure used in our own `baselines_ames.ipynb`, the project is organised as a five-level hierarchy. Each level is meant to answer one specific question, and each level must beat the previous one to justify itself.

Table 1: Project baseline hierarchy. The mid-term covers Levels 0 to 3.

Level	Model class	Representation	Status
0	majority-class / random	none	complete
1	RF, XGBoost	Morgan FP, RDKit 2D	complete
2	ChemBERTa (frozen + head)	SMILES tokens (PLM)	complete (AMES only)
3	GCN, GIN, GAT, GraphSAGE	molecular graph (2D)	complete (v1, v2)
4	SchNet, DimeNet	3D conformer	stretch goal (not started)

This report covers Levels 0 to 3. The class-imbalance interventions (rejection sampling, SMOTE) sit alongside Level 1 as an investigation of why fingerprint baselines lose recall on heavily imbalanced datasets. The final leg extends the strongest pipeline to drug-target interaction (an extension along the dataset axis rather than the representation axis) and, time permitting, attempts Level 4 on LD50.

2 Datasets and Evaluation

2.1 Datasets

We use six toxicity datasets from Therapeutics Data Commons (TDC) [1]. Tox21 is treated as twelve separate single-task endpoints, giving 17 tasks in total (4 binary classification, 1 regression, 12 Tox21 endpoints). Two drug-target interaction datasets (DAVIS and KIBA) are loaded into the same pipeline but reserved for the final leg.

Table 2: Datasets in the project. Positive ratio refers to the toxic (minority) class.

Dataset	Task	Size	Pos. ratio	Endpoint
Toxicity (mid-term)				
AMES	binary classification	7,255	∼50%	Salmonella mutagenicity
hERG_Karim	binary classification	13,445	∼40%	cardiac arrhythmia risk
DILI	binary classification	475	∼45%	drug-induced liver injury
ClinTox	binary classification	1,484	∼8%	clinical-trial toxicity failure
LD50_Zhu	regression	7,385	n/a	acute lethal dose (log scale)
Tox21 (×12)	binary, 12 endpoints	∼7,831	3 to 15%	nuclear-receptor / stress-response panel
Drug-target interaction (final leg)				
DAVIS	regression	30,056 pairs	n/a	68 drugs × 442 kinase targets, K_d
KIBA	regression	118,254 pairs	n/a	2,111 drugs × 229 targets

The collection deliberately spans size (475 to 13,445 molecules), class balance (4% to 50% positive), and task type. ClinTox and the lower-prevalence Tox21 endpoints provide the imbalance stress test, AMES and hERG a balanced reference, DILI a small-data corner case, and LD50 the regression task that strains representation quality.

2.2 Splits and metrics

Classical baselines and the imbalance experiments use scaffold-based train, validation, and test splits, which group molecules by their core chemical skeleton so that test molecules belong to scaffold families unseen during training. This is the standard generalisation test for molecular property prediction. The GNN runs to date use TDC's deterministic random split, and reconciling both pipelines on scaffold splits is the first item in the final-leg plan.

We report ROC-AUC as the primary classification metric (threshold-independent, robust to imbalance), F1 of the positive class as a secondary metric (it reveals whether the model commits to predictions on the minority class), and R^2 and RMSE for LD50 regression. Accuracy is intentionally excluded from comparison: at 4% prevalence on SR-ATAD5, predicting "non-toxic" for everything is 96% accurate but useless. Where decision thresholds are used (for F1), we report both the default 0.5 cut-off and a calibrated threshold equal to the dataset's positive-class ratio.

3 Methods

3.1 Molecular representations

Four input representations are compared. **Morgan fingerprints (ECFP4)** encode each molecule as a 1024-bit vector via RDKit with radius 2 [2, 3], where each bit flags a circular substructure pattern. This is the QSAR default and the baseline any new method must beat. **RDKit 2D descriptors** are a small set of hand-engineered global properties (molecular weight, LogP, TPSA, hydrogen-bond donor and acceptor counts) that medicinal chemists routinely inspect but that lose all sub-structural detail. **ChemBERTa** is a RoBERTa backbone pre-trained on 77 million SMILES strings, used here with the backbone frozen and a small classification head trained on the toxicity label, to test whether large-scale unsupervised pre-training transfers downstream. **Molecular graphs** are produced by parsing each SMILES into a PyTorch Geometric `Data` object via RDKit [8], with atoms as nodes and bonds as edges in both directions. Atom features are described per GNN configuration in §3.3, and bond-level features are not yet encoded (the edge index carries connectivity only).

3.2 Classical baselines

Random Forest (scikit-learn, 500 trees, `min_samples_leaf`=2) and XGBoost (500 estimators, depth 6, learning rate 0.1) are trained on Morgan fingerprints for every classification dataset, on the same input for LD50 regression, and additionally on RDKit 2D descriptors on AMES for the representation comparison.

3.3 GNN architectures

All four architectures share an identical skeleton: two convolutional layers, ReLU activation, global pooling over each graph, and a linear output head. Any performance difference is therefore attributable to the aggregation rule. References: GCN [4], GIN [5], GAT [6], GraphSAGE [7].

Table 3: GNN architectures compared.

Model	Aggregation	Pooling	Theoretical property
GCN	degree-normalised mean	mean	symmetric Laplacian smoothing
GIN	learnable $(1+\varepsilon)$-sum + MLP	sum	as expressive as 1-WL test
GAT	multi-head attention-weighted mean	mean	learns per-edge importance
GraphSAGE	concat(self, mean(neighbours))	mean	inductive generalisation

We trained each architecture in two configurations. **v1** uses a minimal 4-feature atom encoding (atomic number, degree, formal charge, aromaticity), hidden dimension 16, learning rate 0.01, and unweighted binary cross-entropy. **v2** expands the atom encoding to 9 features (adding hydrogen count, ring membership, hybridisation, total valence, and radical electrons), increases hidden dimension to 64, lowers the learning rate to 0.001, and replaces the loss with class-weighted BCE: `pos_weight` $= n_{\text{neg}}/n_{\text{pos}}$ computed per dataset from the training split.

4 Results

4.1 Classical baselines (Level 1)

Random Forest and XGBoost on Morgan fingerprints set the reference performance any learned representation must beat.

Table 4: Classical ML baselines on Morgan fingerprints (scaffold splits).

Dataset	**RF**	**XGBoost**
AMES (50% pos)	0.906	0.881
hERG_Karim (40% pos)	0.920	0.897
DILI (45% pos)	0.868	n/a
ClinTox (8% pos)	0.792	0.870
Tox21 (macro, 3 to 15% pos)	0.776	0.803
LD50_Zhu (R^2)	0.583	0.605

On balanced classification, fingerprints reach 0.88 to 0.92 AUROC. On severely imbalanced ClinTox, RF drops to 0.79. On LD50 regression, both methods plateau at $R^2 \approx 0.6$, leaving roughly 40% of the variance in acute toxicity unexplained.

4.2 Class-imbalance interventions

Because over-prediction of the majority class is the most visible failure mode on toxicity data, we tested two standard interventions before moving to GNNs.

Cost-proportionate rejection sampling. A custom meta-estimator computes a per-batch positive-class weight and probabilistically rejects majority-class samples. Under 5-fold cross-validation, mean ROC-AUC dropped to 0.794 on ClinTox and 0.744 on Tox21, a noticeable regression from the unmodified baselines. The cause is information loss: rejecting thousands of non-toxic molecules to match a tiny positive set throws away most of what the trees were learning from.

SMOTE oversampling. XGBoost with SMOTE-balanced training data, evaluated by per-class precision, recall, and F1 under 5-fold CV. Figure 1 shows the resulting confusion matrices. On Tox21 the model recovered useful minority-class behaviour (positive-class recall 0.46, F1 0.41). On ClinTox the gain was marginal (recall 0.20, F1 0.20). The asymmetry is informative. When fingerprints already separate the classes well (Tox21, large absolute number of positives), SMOTE helps the decision boundary. When fingerprints lack separating information (ClinTox, only ∼119 positives), no resampling can manufacture it.

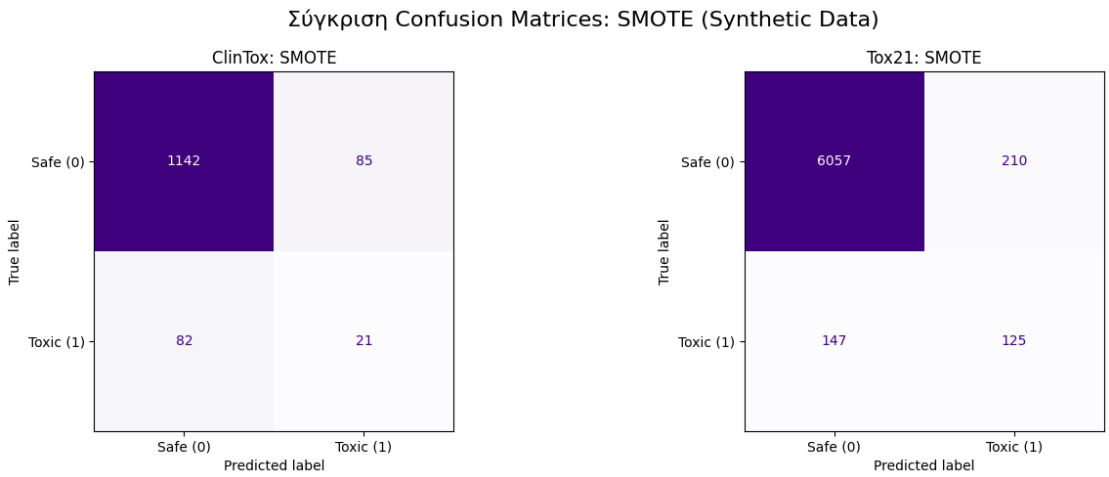

Figure 1: Confusion matrices after SMOTE oversampling on the classical XGBoost pipeline. Tox21 (right) recovers a meaningful minority-class signal (125 of 272 toxic captured). ClinTox (left) does not (21 of 103). SMOTE cannot manufacture separating information that the input representation lacks.

These results establish the baseline lesson: *imbalance is best addressed inside the loss function, not by mutilating the training set.* This directly motivated the class-weighted-loss choice for GNN v2 (§3.3).

4.3 Representation comparison on AMES (Level 2)

AMES (balanced, 50% positive, 7,255 molecules) provides a clean head-to-head between three input representations: Random Forest on Morgan FP and on RDKit 2D descriptors, plus ChemBERTa with frozen backbone and a trainable head.

Table 5: Representation comparison on AMES (scaffold split).

Representation	**Model**	**AUROC**	**F1**	**Accuracy**
Morgan FP (1024-bit)	RF	0.900	0.839	0.825
Morgan FP (1024-bit)	XGBoost	0.899	0.842	0.826
RDKit 2D descriptors	RF	0.867	0.806	0.786
ChemBERTa (77M pre-trained)	frozen + head	0.802	0.750	0.725

Morgan fingerprints win on a balanced, data-rich task. RDKit descriptors trail by 3 AUROC points and the frozen ChemBERTa by 10. Learned embeddings are not automatically better than well-engineered fingerprints, at least not on this regime.

4.4 GNN results, v1 to v2 (Level 3)

The four architectures were trained in two configurations across the 17 tasks (68 runs per configuration). v1 used a minimal 4-feature atom encoding, hidden dimension 16, unweighted BCE, and a fixed 0.5 threshold. v2 expanded atom features to 9 dimensions, raised hidden dimension to 64, and used class-weighted BCE with per-dataset `pos_weight`.

Table 6: GNN v1 vs. v2 on the headline metrics. GIN is strongest in both configurations.

	Tox21 macro-AUROC		**Mean F1 (16 cls.)**		**LD50 R^2**	
Model	v1	v2	v1	v2	v1	v2
GCN	0.705	0.732	0.108	0.279	0.260	0.335
GIN	0.753	**0.781**	0.307	**0.388**	0.430	**0.491**
GAT	0.721	0.748	0.122	0.314	0.206	0.396
GraphSAGE	0.734	0.710	0.187	0.314	0.236	0.415

v1 surfaced two failure modes. LD50 regression was under-fit across all four ($R^2 \leq 0.43$), pointing at insufficient atom information and capacity. F1 collapsed on imbalanced classification: GCN and GAT produced F1 = 0 at the default threshold despite reasonable AUROC. The clearest case was ClinTox/GIN, ROC-AUC 0.922 with F1 = 0, where the model was ranking molecules near-perfectly but never crossing the sigmoid threshold. Calibrating the threshold to the class prior recovered F1 to 0.34 with no retraining.

v2 fixes both failures. Three of four architectures improve on every metric, and GIN remains strongest (Tox21 macro 0.781, 11 of 16 wins, LD50 $R^2 = 0.491$). The class-weighted loss is the largest single contributor to F1 recovery (GAT mean F1 jumps from 0.122 to 0.314). GraphSAGE regresses on three highly imbalanced datasets (ClinTox, NR-ER-LBD, NR-Aromatase) under aggressive `pos_weight`, which focal loss [9] addresses.

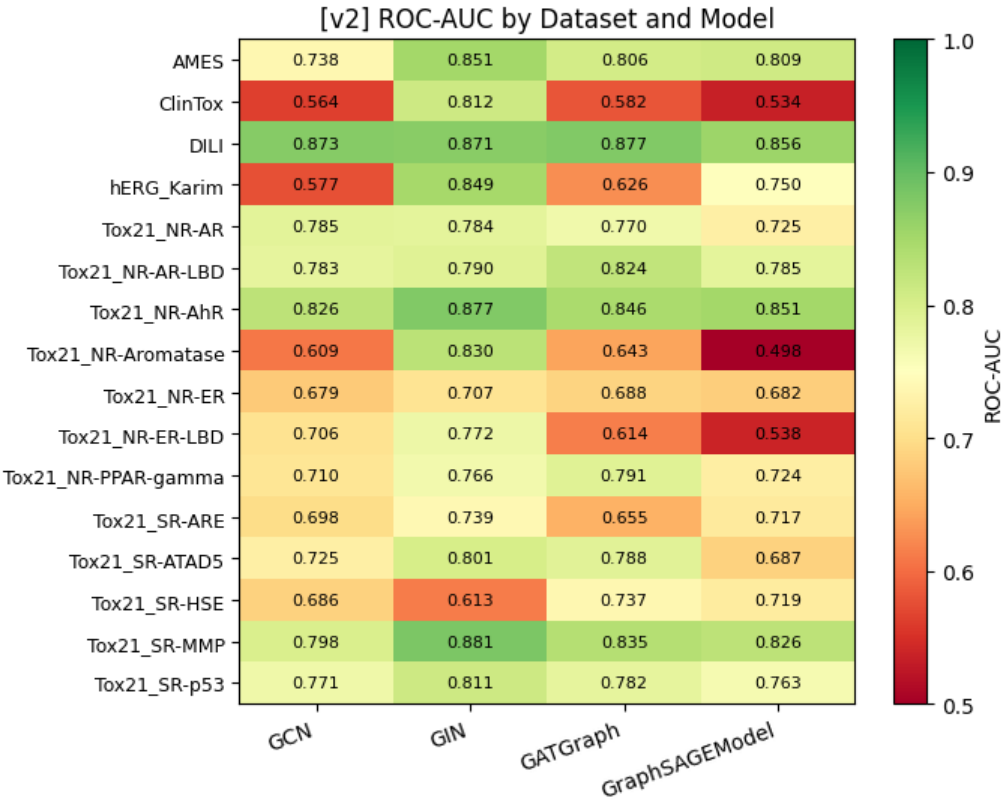

Figure 2: Per-dataset, per-architecture v2 ROC-AUC. GIN is the most uniformly green column. GraphSAGE collapses on the most imbalanced rows (ClinTox 0.534, NR-Aromatase 0.498, NR-ER-LBD 0.538).

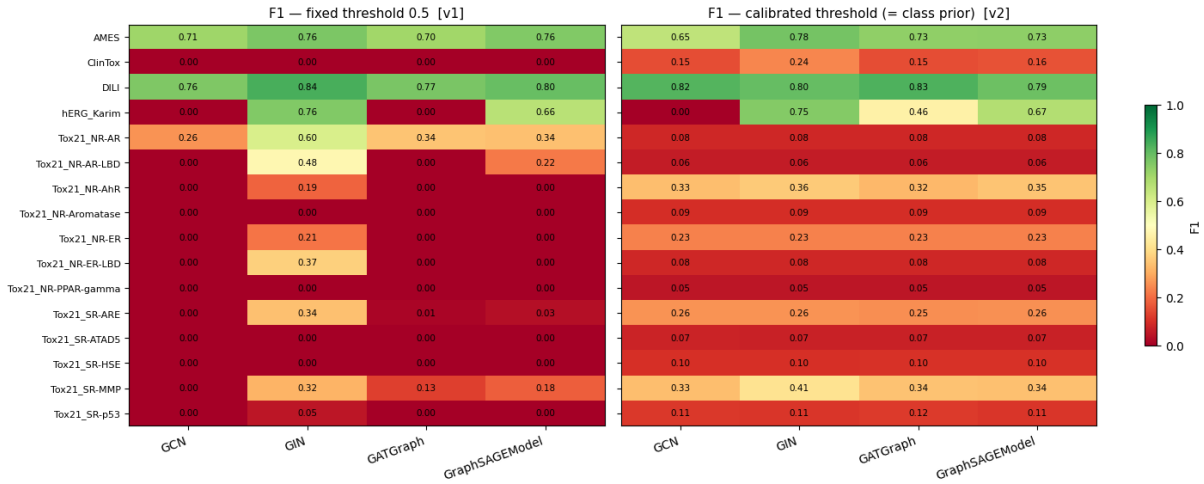

Figure 3: Per-dataset F1 heatmaps. *Left*: v1, fixed 0.5 threshold (red columns are F1 collapse). *Right*: v2, calibrated threshold equal to the class prior. v2 is uniformly non-zero across imbalanced rows, including endpoints that v1 left dead.

Figure 2 shows v2 AUROC mostly above 0.7 across endpoints with GIN best. Figure 3 shows the threshold-calibration effect: v2 trades a small amount of peak F1 on the balanced rows for non-zero F1 across every imbalanced row, the signature of a model taught to commit to the minority class during training rather than rescued at evaluation.

4.5 GNN vs. classical head-to-head

Bringing the strongest GNN (GIN v2) head-to-head with the strongest classical baseline per task gives the cleanest answer to the proposal's core question on the toxicity stack.

Table 7: Best classical ML vs. GIN v2 on the comparable subset of tasks. Bold marks the winner per row. Ties within 0.01 AUROC are not bolded.

Task	**Best classical**	**GIN v2**	**Verdict**
AMES (50% pos)	**0.906** (RF)	0.851	classical wins by 0.06
hERG_Karim (40% pos)	**0.920** (RF)	0.849	classical wins by 0.07
DILI (45% pos, n=475)	0.868 (RF)	0.877 (GAT v2)	tie / GAT slightly ahead
ClinTox (8% pos)	0.870 (XGB)	0.812	XGB wins, GIN beats RF (0.792)
Tox21 (macro)	**0.803** (XGB)	0.781	classical wins by 0.02
LD50_Zhu (R^2)	**0.605** (XGB)	0.491	classical wins by 0.11

The picture is more nuanced than either "fingerprints win" or "GNNs win". On balanced classification (AMES, hERG), where fingerprints have abundant data and the substructure patterns are highly informative, classical ML keeps a clear edge. On the small DILI dataset, GAT v2 edges ahead of RF. That is the only outright GNN win. On ClinTox, GIN v2 beats RF but loses to XGBoost. On Tox21 the gap closes to 0.02 AUROC. On LD50 regression, classical ML retains a substantial 11-point R^2 advantage.

5 Discussion

5.1 Where the fingerprint baseline is weak

Across the toxicity stack, classical ML on Morgan fingerprints loses (or comes close to losing) only on small data (DILI, n=475) and on severe class imbalance (ClinTox 8%). Where it wins comfortably (AMES, hERG, LD50), the dataset is large or balanced or both. The frozen ChemBERTa on AMES tightens the picture: a 77M-parameter chemical language model loses to a 1024-bit fingerprint plus 500 trees by 10 AUROC points. Pre-trained learned embeddings do not automatically transfer.

5.2 Loss-side vs. data-side imbalance fixes

Three independent results on the imbalance question converge:

- Rejection sampling on classical ML *reduces* AUROC (information loss).
- SMOTE on classical ML helps when fingerprints already separate the classes (Tox21) and is ineffective when they do not (ClinTox).
- Class-weighted BCE inside the GNN loss roughly triples mean F1 on the architectures that had collapsed, with no data manipulation.

The lesson generalises beyond this project: imbalance interventions should reshape the gradient signal, not the dataset. The picture of v2 in Figure 3 (right) is the picture of a model that learned to commit to the minority class during training, not one rescued at evaluation time by lowering a threshold.

5.3 Limitations

Inconsistent splits across pipelines.
Classical baselines and the SMOTE / rejection-sampling experiments use scaffold splits. The GNN runs use TDC's deterministic random split. The head-to-head table above is therefore an indication, not yet a fair comparison, and re-running the GNNs on scaffold splits is the first item in the next-steps plan.

Single seed.
Each configuration was trained once. We have no confidence intervals. Differences smaller than roughly 2 AUROC points should not be over-interpreted.

No bond features in the GNNs.
Bonds carry only connectivity, not type, stereochemistry, or ring membership at the edge level. This is the largest within-graph information gap.

ChemBERTa tested on one dataset only.
We do not know whether learned embeddings fail under imbalance the same way the v1 GNNs did.

No DTI experiments yet, no 3D yet.
Levels 4 (3D) and the dataset-axis extension (DTI) are not started.

6 Final Phase

6.1 Research question

The proposal's question is answered narrowly on toxicity: GIN matches fingerprints only on small-data and severely imbalanced regimes, and trails on the rest. Pushing further along the

representation axis (more features, 3D) would explore the same regime where the engineered baseline is already strong. The decisive test is drug-target interaction, where the protein partner cannot be Morgan-fingerprinted and the GNN drug encoder is paired with a protein encoder by construction. A fingerprint baseline must improvise on the protein side, while the GNN operates in its native regime. This is where the proposal originally aimed, and where the GNN's structural advantage should be largest.

> *Where does the graph neural network provide a meaningful advantage over Morgan-fingerprint baselines on drug safety prediction? On drug-only toxicity the answer is narrow: GIN matches fingerprints only on small or severely imbalanced regimes. On drug-target interaction, where the protein partner is not amenable to fingerprinting and a GNN drug encoder is paired with a protein encoder by design, does the GNN's structural advantage become decisive?*

The mid-term answers the first half. The final phase answers the second.

6.2 Plan

The final phase hardens the GNN we already trained and takes it to drug-target interaction. The two steps are sequential. The DTI step is the one that answers the research question.

1. **Harden the GNN on toxicity.** Three changes targeting known weaknesses of v2. First, add bond-level edge features (type, stereochemistry, ring membership), the largest remaining within-graph information gap. Second, re-run all GNN configurations on scaffold splits with three seeds to give Table 7 confidence intervals. Third, replace GraphSAGE's aggressive `pos_weight` with focal loss [9] to recover its three regressed datasets. Output: a rigorous, scaffold-split, multi-seed head-to-head against the classical baseline on the same toxicity tasks.

2. **Apply the hardened GNN to DAVIS.** DAVIS is already loaded (30k drug-target pairs, 68 drugs $\times$ 442 kinase targets, K_d regression). The GIN drug encoder from step 1 is paired with a frozen ESM-2 small protein encoder and a linear projection head on the target side, concatenated and passed through an MLP for the final scalar. The fingerprint baseline gets a Morgan fingerprint on the drug side and an amino-acid-composition embedding on the protein side. Metrics: Pearson, RMSE, concordance index. The win condition is whether the GNN drug encoder beats the fingerprint drug encoder when both are paired with the same protein representation, and by how much.

3. **Final analysis and report.** Combined comparison across toxicity and DTI on consistent splits with confidence intervals.

6.3 Stretch goals

These run only if the DTI experiment converges with time remaining. They do not hold up the final report.

Level 4 (SchNet on LD50).
Train SchNet [10] on RDKit-generated 3D conformers for LD50. The cleanest place to test whether 3D atomic geometry adds anything that 2D plus bond features did not.

KIBA.
The larger DTI dataset (118k pairs). Attempted only if DAVIS is solved decisively.

Software

The pipeline is built on standard open-source libraries: RDKit for SMILES parsing and atom or bond featurisation, PyTorch and PyTorch Geometric [8] for the GNNs, scikit-learn and XGBoost for the classical baselines, HuggingFace Transformers for ChemBERTa, and TDC [1] for dataset access and scaffold splits. All experiments use seed 42 across `random`, `numpy`, `torch`, and CUDA. Per-version model weights, prediction CSVs, and TensorBoard logs are saved under `models/v{1,2}/` and `results/v{1,2}/` so each phase remains reproducible alongside later experiments.

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